INCH-POUND

MIL-S-20708/62F
13 March 1989

SUPERSEDING
MIL-S-20708/62E
30 July 1987

MILITARY SPECIFICATION SHEET
SYNCHRO, TORQUE RECEIVER TRANSMITTER, TYPE 31TRX4A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Synchros described herein shall consist of this specification and the latest issue of MIL-S-20708.

TABLE I Requirements

Requirement	Value	Unit	Tolerance
Frequency	400	Hz	±1%
Primary Voltage	115	volts	±1%
Primary Current	1650 0	milliamps	maximum
Primary Power	14 4	watts	maximum
Impedance			
Zss	2 5-5 5	ohms	min -max
Impedance Angle			
Zss	60 0-75 0	degrees	min -max
Transformation Ratio	0 783	-----	±2%
Phase Shift (Lead)	2 0	degrees	maximum
Electrical Error	8.0	minutes	maximum
Receiver Error	36 0	minutes	maximum
Null Voltage:			
Total	170 0	millivolts	maximum
Fundamental	35 0	millivolts	maximum
Synchronizing Time			
30° ±2°	1 0	seconds	maximum
Synchronizing Time			
177° ±2°	2 0	seconds	maximum
Torque Gradient	0 67	ounce-inches per degree	minimum
Radial Play	0 0008	inches	maximum
End Play	0 0030	inches	±0 0020
Temperature Rise	60 0	degrees (C)	maximum
Variation of Voltage (+10%) and Freq (-5%)	27 0	watts	maximum

This unit shall not be subjected to the altitude requirements of MIL-S-20708.

(F) denotes changes

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TABLE II Military part number variant characteristics

Military Part No 1/	A ± 010	B ± 010	G Maximum	Shaft Style	Terminal End	U + 0000 − 0002
M20708/62-01A	700	730	3 380	Splined	Terminal Board	2405

(F)

1/Military part number M20708/62-01A shall be the NATO standard

1/The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation

TABLE III Military part number cross-references

SUPERSEDED MILITARY PART NUMBERS		NEW MILITARY PART NUMBERS
MIL-S-20708/62D	MIL-S-20708/62C	
M20708/62D-001	M20708/62-001	M20708/62-01A

NOTE All line items shown in Table III refer to equivalent and interchangeable synchros of the same type designation modification. Part number changes do not affect form, fit, or function of the synchros listed therein

Custodians
Navy-AS

Review Activities
Army-EC
DLA-ES

User Activities
Navy-MC CG

Preparing Activity
Navy-AS

(Project 5990-0349-39)

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